# Timing a possible encounter

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Separate orbital period from an encounter time.

**SUCCESS CRITERIA - I can...**

1. Use a clearly stated period and time reference.
2. Check positions of both bodies at the same time.
3. Explain why geometry and uncertainty matter.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

At fixed mass and speed, circle radius becomes 3 times as large. Compare acceleration and net force using equations.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $t_{arrival} = t_{start}$ + travel time.
An encounter needs matching position and time. A period T is one complete orbit. Use the supplied fictional timing model.

**Worked reasoning:** Two objects pass the same crossing point: A at months 2, 8, 14 and B at months 5, 14, 23. Only month 14 is a shared time in these lists.

**Your turn:** Draw two timelines and connect the matching time. Explain why a shared location alone is insufficient.

**Marzano check - circle:** ① ② ③

## 03 Represent and interpret

Object A reaches a crossing in 6 months. Earth reaches it in 5 months and again 15 months from now. Does crossing the same point establish an encounter?

| Body | Crossing time |
|---|---|
| A | 6 months |
| Earth | 3, 15 months |

**Marzano check - circle:** ① ② ③

## 04 Solve and justify

Toy crossings: asteroid at months 6, 24, 42; Earth at months 6, 18, 30, 42. Identify common crossing times and explain a limit of the model.

**Marzano check - circle:** ① ② ③

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Continue down each column.

## 05 Analyze the reasoning

A student turns a ±2-day timing range into “a 50% chance of impact.” Explain why that does not follow.

**Marzano check - circle:** 1 2 3

## 06 Choose a method

An object weighs 39.2 N where g=9.8 N/kg. Find its mass and its weight in a field of 6 N/kg.

**Marzano check - circle:** 1 2 3

## 07 Exit check

An asteroid’s crossing window is days 10-14; Earth’s is days 13-15. Find overlap and state what it does and does not establish.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: Apply it another way

A reaches a crossing at months 1+6 n; B at months 3+8 m, for nonnegative integers n, m. Find their first common crossing time and a limit of this timing model.

**Marzano check - circle:** 1 2 3